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ФАКУЛЬТЕТ _____ «Информатика и системы управления»

КАФЕДРА _____ «Теоретическая информатика и компьютерные технологии»

Лабораторная работа № 4
по курсу «Разработка параллельных и распределенных
программ»
«Задача о пяти обедающих философах»

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Москва 2025

1 Задача

Пять философов сидят за круглым столом. Они проводят жизнь, чередуя приёмы пищи и размышления. В центра стола находится большое блюдо спагетти. Чтобы съесть порцию, каждому философу нужно две вилки. Однако, вилок всего пять: между каждой парой рядом сидящих философов лежат по одной вилке, и каждый философ может пользоваться только теми вилками, которые лежат рядом с ним, слева и справа. Философ не может брать две вилки одновременно: сначала он тратит некоторое время на то, чтобы взять одну, затем вторую. Однако, он может одновременно положить их на место.

Задача заключается в том, чтобы написать программу, моделирующую поведение философов. Очевидно, что раз вилок всего пять, то одновременно есть могут не более двух философов, и два сидящих рядом философа не могут есть одновременно. Для имитации периодов раздумий и приёмов пищи можно использовать генератор случайных чисел, позволяющий задавать времена их действий в определённом интервале. Имитация поведения каждого философа, по сути, разбивается на то, что в любой момент времени философ находится в одном из пяти состояний: размышляет, берёт левую вилку, берёт правую вилку, ест, кладёт вилки на место. Таким образом, вилки являются разделяемым ресурсом.

На программу накладываются условия:

1. Каждый философ, по сути, является потоком, и модель поведения у каждого из них должна быть одинаковой, кроме того, какие вилки они могут брать.
2. Накладывание блокировки по сути является действием по взятию вилки, поэтому накладывать блокировку сразу на обе вилки нельзя; последовательность действий должна быть «наложить блокировку – взять вилку – наложить вторую блокировку – взять вторую вилку».
3. Программа должна избегать ситуации взаимоблокировки: ситуации, в которой все философы голодны, то есть ни один из них не может взять себе две вилки (например, когда каждый держит по одной и не хочет её отдавать).

2 Практическая реализация

Исходный код программы представлен в листингах 1– 2– ??.

Листинг 1 — часть 1

```

1 package main
2 import (
3     "flag"
4     "fmt"
5     "math/rand"
6     "sync"
7     "time"
8 )
9 const (
10     stateThinking = "is thinking"
11     stateTakeLeft  = "takes left fork"
12     stateTakeRight = "takes right fork"
13     stateEating    = "is eating"
14     statePutForks  = "puts down forks"
15 )
16 type Philosopher struct {
17     id          int
18     leftFork    *sync.Mutex
19     rightFork   *sync.Mutex
20 }
21 type Logger struct {
22     mu sync.Mutex
23 }
24 func (l *Logger) Log(id int, state string) {
25     l.mu.Lock()
26     defer l.mu.Unlock()
27     fmt.Printf("%v: philosopher %d %s\n", time.Since(startTime).Truncate(
28         time.Millisecond), id, state)
29 }
30 var startTime time.Time
31 func (p *Philosopher) Run(stop <-chan struct{}, logger *Logger, minThink
32     , maxThink, minEat, maxEat time.Duration, leftFirst bool) {
33     r := rand.New(rand.NewSource(time.Now().UnixNano() + int64(p.id)))
34     for {
35         select {
36             case <-stop: return
37             default:
38                 logger.Log(p.id, stateThinking)
39                 thinkTime := minThink + time.Duration(r.Int63n(int64(maxThink-
40                     minThink)))
41                 time.Sleep(thinkTime)
42                 if leftFirst {
43                     p.leftFork.Lock()
44                     logger.Log(p.id, stateTakeLeft)
45                     p.rightFork.Lock()
46                     logger.Log(p.id, stateTakeRight)
47                 } else {
48                     p.rightFork.Lock()
49                     logger.Log(p.id, stateTakeRight)
50                     p.leftFork.Lock()
51                     logger.Log(p.id, stateTakeLeft)
52                 }
53                 logger.Log(p.id, stateEating)
54                 eatTime := minEat + time.Duration(r.Int63n(int64(maxEat-minEat)))
55                 time.Sleep(eatTime)
56                 logger.Log(p.id, statePutForks)
57                 p.leftFork.Unlock()
58                 p.rightFork.Unlock()
59             }
60     }
61 }

```

Листинг 2 — часть 2

```
1 func main() {
2     var (
3         nPhilosophers = flag.Int("n", 5, "number of philosophers")
4         totalSeconds   = flag.Int("t", 10, "simulation duration in seconds")
5     )
6     flag.Parse()
7
8     fmt.Printf("Dining Philosophers Problem: N=%d, simulation time=%d\n", *nPhilosophers, *totalSeconds)
9
10    startTime = time.Now()
11
12    forks := make([]*sync.Mutex, *nPhilosophers)
13    for i := 0; i < *nPhilosophers; i++ {
14        forks[i] = &sync.Mutex{}
15    }
16
17    logger := &Logger{}
18    stop := make(chan struct{})
19    var wg sync.WaitGroup
20
21    minThink := 200 * time.Millisecond
22    maxThink := 800 * time.Millisecond
23    minEat := 200 * time.Millisecond
24    maxEat := 800 * time.Millisecond
25
26    for i := 0; i < *nPhilosophers; i++ {
27        left := forks[i]
28        right := forks[(i+1)%*nPhilosophers]
29        p := &Philosopher{id: i, leftFork: left, rightFork: right}
30
31        wg.Add(1)
32        leftFirst := (i%2 == 0)
33
34        go func(ph *Philosopher, lf bool) {
35            defer wg.Done()
36            ph.Run(stop, logger, minThink, maxThink, minEat, maxEat, lf)
37        }(p, leftFirst)
38    }
39
40    time.Sleep(time.Duration(*totalSeconds) * time.Second)
41    close(stop)
42    wg.Wait()
43
44    fmt.Println("Simulation completed.")
45 }
```

3 Характеристики устройства

Модель оборудования: MEGABOOK T14TA (TECNO T14TA)

Процессор: 11th Gen Intel(R) Core(TM) i5-115

Память: 16 ГиБ

Емкость диска: 512 ГБ

4 Лог программы

Dining Philosophers Problem: N=5, simulation time=10 seconds

0s: philosopher 4 is thinking
0s: philosopher 1 is thinking
0s: philosopher 0 is thinking
0s: philosopher 3 is thinking
0s: philosopher 2 is thinking
238ms: philosopher 1 takes right fork
238ms: philosopher 1 takes left fork
238ms: philosopher 1 is eating
471ms: philosopher 4 takes left fork
471ms: philosopher 4 takes right fork
471ms: philosopher 4 is eating
830ms: philosopher 4 puts down forks
830ms: philosopher 4 is thinking
830ms: philosopher 0 takes left fork
830ms: philosopher 3 takes right fork
830ms: philosopher 3 takes left fork
830ms: philosopher 3 is eating
958ms: philosopher 1 puts down forks
958ms: philosopher 1 is thinking
958ms: philosopher 2 takes left fork
958ms: philosopher 0 takes right fork
959ms: philosopher 0 is eating
1.052s: philosopher 3 puts down forks
1.052s: philosopher 2 takes right fork
1.052s: philosopher 2 is eating
1.052s: philosopher 3 is thinking
1.25s: philosopher 4 takes left fork
1.34s: philosopher 0 puts down forks
1.34s: philosopher 0 is thinking
1.34s: philosopher 4 takes right fork

1.34s: philosopher 4 is eating
1.437s: philosopher 2 puts down forks
1.437s: philosopher 2 is thinking
1.437s: philosopher 1 takes right fork
1.437s: philosopher 1 takes left fork
1.437s: philosopher 1 is eating
2.118s: philosopher 4 puts down forks
2.118s: philosopher 4 is thinking
2.118s: philosopher 0 takes left fork
2.118s: philosopher 3 takes right fork
2.118s: philosopher 3 takes left fork
2.118s: philosopher 3 is eating
2.229s: philosopher 1 puts down forks
2.229s: philosopher 1 is thinking
2.229s: philosopher 2 takes left fork
2.229s: philosopher 0 takes right fork
2.229s: philosopher 0 is eating
2.653s: philosopher 3 puts down forks
2.653s: philosopher 3 is thinking
2.653s: philosopher 4 takes left fork
2.653s: philosopher 2 takes right fork
2.653s: philosopher 2 is eating
2.869s: philosopher 0 puts down forks
2.869s: philosopher 0 is thinking
2.869s: philosopher 4 takes right fork
2.869s: philosopher 4 is eating
3.183s: philosopher 2 puts down forks
3.183s: philosopher 2 is thinking
3.183s: philosopher 1 takes right fork
3.183s: philosopher 1 takes left fork
3.183s: philosopher 1 is eating
3.483s: philosopher 4 puts down forks
3.483s: philosopher 4 is thinking

3.483s: philosopher 0 takes left fork
3.483s: philosopher 3 takes right fork
3.483s: philosopher 3 takes left fork
3.483s: philosopher 3 is eating
3.611s: philosopher 1 puts down forks
3.611s: philosopher 1 is thinking
3.611s: philosopher 0 takes right fork
3.611s: philosopher 0 is eating
3.877s: philosopher 2 takes left fork
4.08s: philosopher 3 puts down forks
4.08s: philosopher 3 is thinking
4.08s: philosopher 4 takes left fork
4.08s: philosopher 2 takes right fork
4.08s: philosopher 2 is eating
4.281s: philosopher 0 puts down forks
4.281s: philosopher 0 is thinking
4.281s: philosopher 4 takes right fork
4.281s: philosopher 4 is eating
4.582s: philosopher 2 puts down forks
4.582s: philosopher 2 is thinking
4.582s: philosopher 1 takes right fork
4.582s: philosopher 1 takes left fork
4.582s: philosopher 1 is eating
5.009s: philosopher 4 puts down forks
5.009s: philosopher 0 takes left fork
5.009s: philosopher 3 takes right fork
5.009s: philosopher 3 takes left fork
5.009s: philosopher 3 is eating
5.009s: philosopher 4 is thinking
5.098s: philosopher 1 puts down forks
5.098s: philosopher 1 is thinking
5.098s: philosopher 2 takes left fork
5.098s: philosopher 0 takes right fork

5.098s: philosopher 0 is eating
5.559s: philosopher 0 puts down forks
5.559s: philosopher 0 is thinking
5.752s: philosopher 3 puts down forks
5.753s: philosopher 3 is thinking
5.753s: philosopher 4 takes left fork
5.753s: philosopher 4 takes right fork
5.753s: philosopher 4 is eating
5.753s: philosopher 2 takes right fork
5.753s: philosopher 2 is eating
6.273s: philosopher 2 puts down forks
6.273s: philosopher 2 is thinking
6.273s: philosopher 1 takes right fork
6.273s: philosopher 1 takes left fork
6.273s: philosopher 1 is eating
6.361s: philosopher 4 puts down forks
6.361s: philosopher 4 is thinking
6.361s: philosopher 0 takes left fork
6.361s: philosopher 3 takes right fork
6.361s: philosopher 3 takes left fork
6.361s: philosopher 3 is eating
6.748s: philosopher 1 puts down forks
6.748s: philosopher 1 is thinking
6.748s: philosopher 2 takes left fork
6.748s: philosopher 0 takes right fork
6.748s: philosopher 0 is eating
6.886s: philosopher 3 puts down forks
6.886s: philosopher 3 is thinking
6.886s: philosopher 4 takes left fork
6.886s: philosopher 2 takes right fork
6.886s: philosopher 2 is eating
7.379s: philosopher 0 puts down forks
7.379s: philosopher 0 is thinking

7.379s: philosopher 4 takes right fork
7.379s: philosopher 4 is eating
7.684s: philosopher 2 puts down forks
7.684s: philosopher 2 is thinking
7.684s: philosopher 1 takes right fork
7.684s: philosopher 1 takes left fork
7.684s: philosopher 1 is eating
7.972s: philosopher 4 puts down forks
7.972s: philosopher 4 is thinking
7.972s: philosopher 3 takes right fork
7.972s: philosopher 3 takes left fork
7.972s: philosopher 3 is eating
7.996s: philosopher 0 takes left fork
8.364s: philosopher 3 puts down forks
8.364s: philosopher 3 is thinking
8.416s: philosopher 1 puts down forks
8.416s: philosopher 1 is thinking
8.416s: philosopher 2 takes left fork
8.416s: philosopher 2 takes right fork
8.416s: philosopher 2 is eating
8.416s: philosopher 0 takes right fork
8.416s: philosopher 0 is eating
8.561s: philosopher 4 takes left fork
8.763s: philosopher 2 puts down forks
8.763s: philosopher 2 is thinking
8.763s: philosopher 1 takes right fork
8.865s: philosopher 0 puts down forks
8.865s: philosopher 0 is thinking
8.865s: philosopher 1 takes left fork
8.865s: philosopher 1 is eating
8.865s: philosopher 4 takes right fork
8.865s: philosopher 4 is eating
9.557s: philosopher 4 puts down forks

9.557s: philosopher 4 is thinking
9.557s: philosopher 0 takes left fork
9.557s: philosopher 3 takes right fork
9.557s: philosopher 3 takes left fork
9.557s: philosopher 3 is eating
9.585s: philosopher 1 puts down forks
9.585s: philosopher 1 is thinking
9.585s: philosopher 2 takes left fork
9.585s: philosopher 0 takes right fork
9.585s: philosopher 0 is eating
10.056s: philosopher 3 puts down forks
10.056s: philosopher 4 takes left fork
10.056s: philosopher 2 takes right fork
10.056s: philosopher 2 is eating
10.097s: philosopher 0 puts down forks
10.097s: philosopher 4 takes right fork
10.097s: philosopher 4 is eating
10.503s: philosopher 4 puts down forks
10.702s: philosopher 2 puts down forks
10.702s: philosopher 1 takes right fork
10.702s: philosopher 1 takes left fork
10.702s: philosopher 1 is eating
11.202s: philosopher 1 puts down forks
Simulation completed.